PANGASINAN STATE UNIVERSITY
Urdaneta City Campus
College of Engineering and Architecture
Mechanical Engineering Department
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PAPER AND PULP INDUSTRY

INDUSTRIAL PLANT ENGINEERING
(PROFME_105)

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INTRODUCTION

Paper is a product manufactured conventionally from wood that has been reduced to individual fibers or fiber bundles (called pulp), formed into a mat from a slurry with water, compressed and dried. **Pulp** is a fibrous material originally derived from wood and non-wood fibers like cotton, sugarcane bagasse, rice straw, and other cellulosic materials by chemical or mechanical processes for use in making paper.

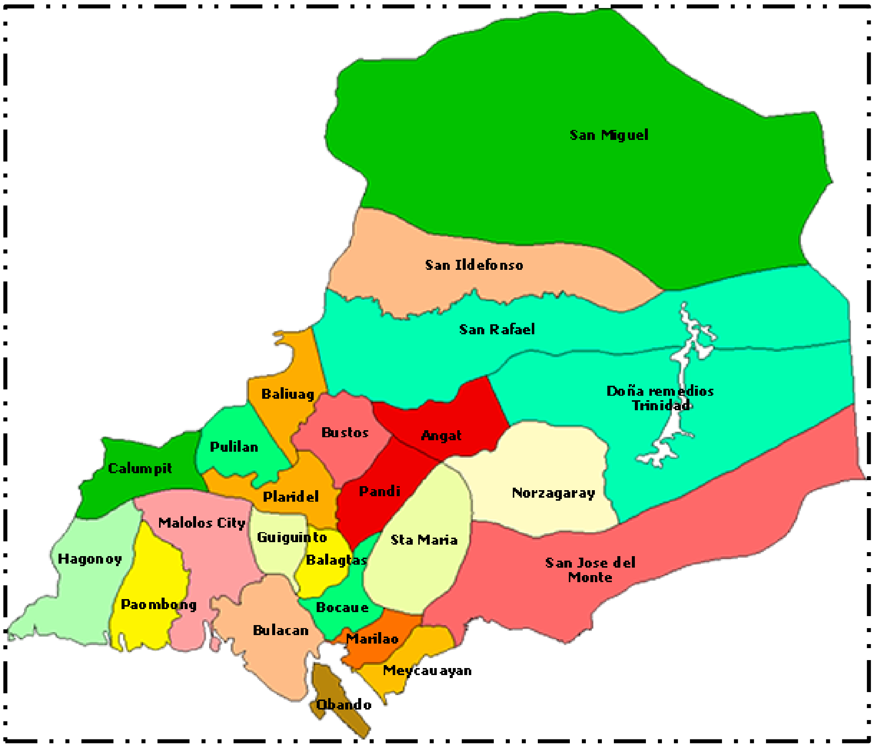
Paper and paper-based products are categorized into different grades, namely: printing and writing, newsprint, corrugating container boards (linerboard and fluting material), tissue and disposable/sanitary papers, other packaging paper and paperboard (bag/wrapping/envelope papers, folding/setup carton boards), and specialty paper and boards (art paper, construction board, etc.)

Raw materials used in paper and pulp industry plays a critical role in determining the quality, texture, and sustainability of the final product. These key materials include wood pulp, non-wood pulp, recycled waste paper, other cellulosic material and various chemicals.

However, due to the lack of virgin wood pulp, the **Philippine paper and pulp industry primarily relies on recycled wastepaper as its main raw material.** Recently, the use of wastepaper accounts for more than half of the fiber raw materials from the viewpoint of energy and resource saving. The paper industry contributes to society as a recycling-oriented industry through the use of wastepaper and the promotion of afforestation projects. Recycled paper pulp is produced by processing used paper products through a process involving deinking and breaking down the fibers or fiber separation. The utilization of recycled materials offers significant environmental advantages, such as waste reduction, conservation of natural resources, and decreased energy consumption and greenhouse gas emissions associated with paper manufacturing. While locally available wastepaper is affordable, its quality and quantity often fall short of meeting the requirements of recycled paper mills. Consequently, these mills supplement their supply by importing wastepaper and virgin pulp to enhance the quality of recycled fibers for paper production, particularly tissue and packaging.

Manufacturing methods for pulp made from wood are roughly classified into the mechanical pulp manufacturing methods in which physical force is applied to wood to mill it, and the chemical pulp manufacturing methods in which extract fibers with chemical agents. Use of wastepaper is roughly classified into gray stock to be used for baseboards for corrugated fiberboard and paperboard for paper boxes, and DIP (deinked pulp) from deinked and bleached wastepaper for use as printing paper and computer paper. The process of preparation and papermaking by appropriately mixing such pulp is used to produce paper of various qualities and features. Considerable energy (electricity and steam) is used in the manufacturing processes of both pulp and paper. Electric power is used to drive motors in machines and facilities, pumps to transport raw materials and wastewater and fan motors, and steam is used to heat and dry raw materials. Paper industry ranks fourth in energy consumption among manufacturing industries, following the iron and steel, chemical, and cement industries. Many paper mills therefore install boilers and turbines and try to improve energy efficiency by introducing cogeneration systems.

PLANT LOCATION

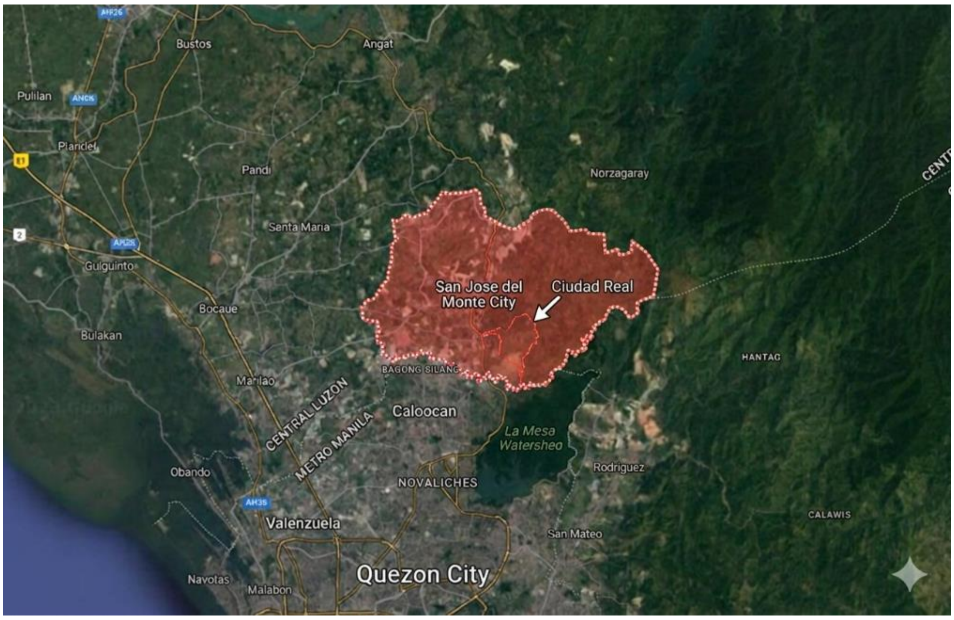


- BACKGROUND OF CHOSEN LOCATION

Bulacan is one of the first-class provinces in the Philippines situated in Central Luzon. It is bounded by the provinces of Aurora and Quezon on the east, Nueva Ecija on the north, Pampanga on the west, Metro Manila Rizal on the south and Manila Bay on the southwest. It has a total land area of 279,610 hectares. The province has 21 municipalities and 3 component cities (Malolos, the capital; Meycauayan and San Jose Del Monte)

San Jose Del Monte (SJDM), a landlocked component city in the province of Bulacan was chosen as location for the construction of a new paper and pulp plant. The economic growth of SJDM had been steadily improving since it was converted into a component city of the province. A strong advantage of SJDM is its location and accessibility. This inland city is approximately only 42 kilometers away from Manila. **Three national roads, four provincial roads, 402 city roads, and 15 bridges**

make SJDM accessible. Moreover, with ongoing infrastructure improvements benefiting the city, SJDM became easy to reach, making it an attractive option for people looking for a place that is not as busy as the metro but close enough to job opportunities in the capital region.



SOME IMPORTANT FACTORS CONSIDERED IN PLANT LOCATION SELECTION

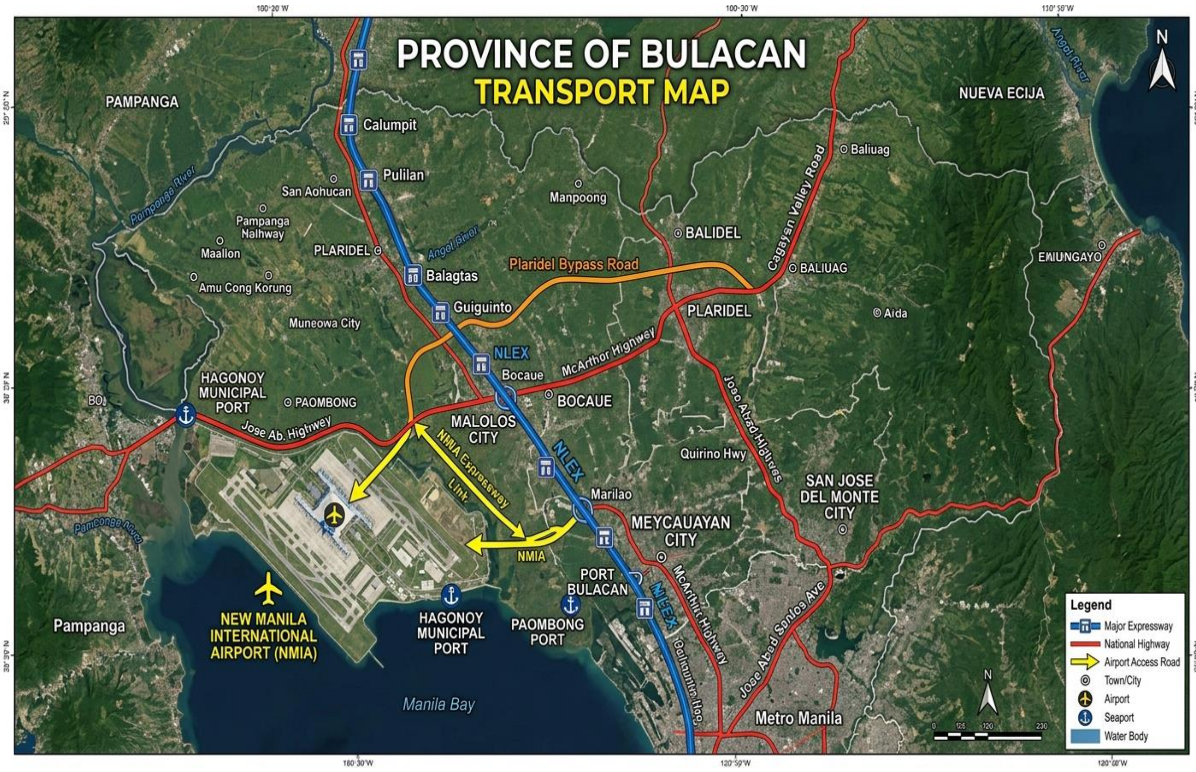
- *RAW MATERIALS SOURCES*

For a quick overview, the only integrated (forestry-to-papermaking operations) pulp and paper mill in the Philippines, PICOP Resources Inc., ceased operations in 2010 due mainly to lack of investment capital to modernize production technologies, and <u>inability to sustain tree plantation program (which led to inadequate supply of wood for its pulp mill).</u> Due to the lack of virgin wood sources, the paper and pulp industry adapted some methods.

Today, the Philippine paper industry sources its fiber material mostly from **recovered wastepaper** that is why <u>the proposed plant is situated near sources of locally-available wastepaper such as in Metro Manila and other urbanized areas</u> that possess high concentration and density in paper consumption, without neglecting possible complementary sources of agricultural wastes which is used as raw materials for paper and pulp manufacturing (such rice straw, bamboo and bagasse) from Region 3. The region is well-known as Central Luzon with its longest contiguous area of lowlands/kapatagan. In fact, the region produces one third of the country's total rice production, thus is also called the Rice Granary of the Philippines.

Waste paper contributes to nineteen percent (19%) of the total municipal solid waste in the Philippines. Metro Manila and other urbanized areas in the country show the highest density in paper consumption. The continuous increase in waste paper generation would potentially provide enough raw materials for paper millers. Thus, there is a great potential for recycling of wastepaper.

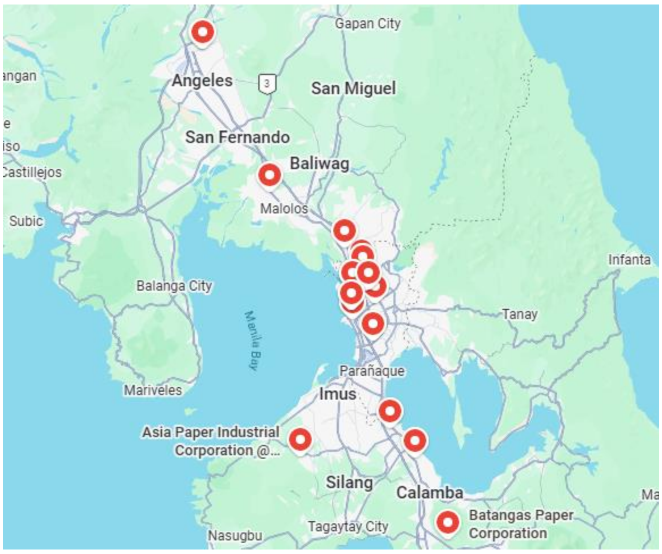
However, the quality and volume of locally-available wastepaper is not sufficient to meet the demands especially in quality. In line with this, Virgin pulp also produce much stronger or high-quality paper grades when blended and used to supplement raw material requirements of paper mills. In developed countries, tree plantations support the wood requirements for pulp and paper production in a renewable and environmentally sustainable manner. **This is not now practiced in the Philippines**. To address the current issue, paper mills resort to importation to help meet their fiber requirements in terms of quantity and quality. This is also why the proposed plant is strategically located near seaports/freight services for convenient and cost-effective importation of raw materials.



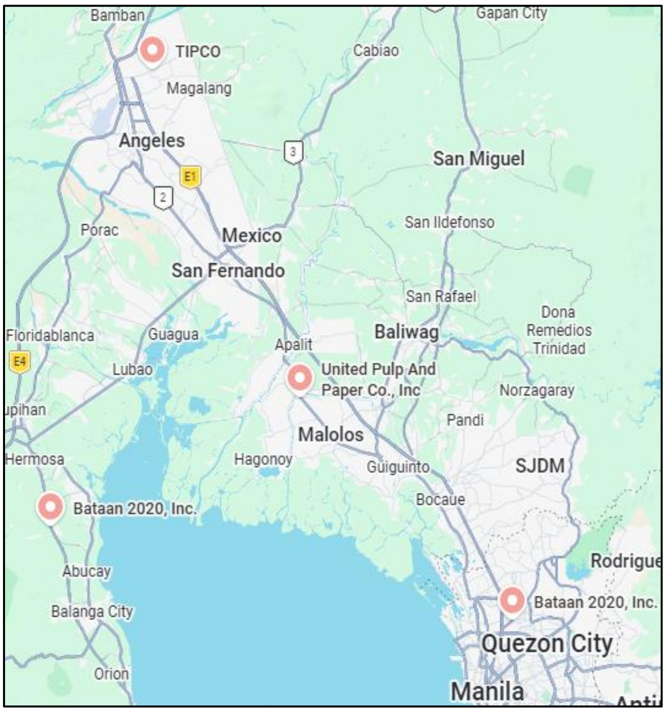
- *TRANSPORTATION AND ACCESS*

Bulacan is one of the few provinces in the Philippines that greatly benefits from its geographic location. It boasts of its <u>strategic location wherein it lies equidistant with the northern and southern parts of Luzon</u>. At the same time, it is <u>proximate and accessible to the National Capital region (NCR) or Metro Manila where most of the development impulses originate</u>. Dubbed as the “Gateway to the North", Bulacan links Metro Manila to the resource-rich provinces of Central and Northern Luzon. The province accessibility to major airports such as NAIA, Clark, Subic as well as piers and Dingalan Bay, and extensive road networks, provides logistical advantage to investors. The province also has modern telecommunications facilities, ensuring reliable communication services.

- *EXISTING HIGH CAPACITY AND FULLY OPERATIONAL PAPER MILLS SURROUNDING THE CHOSEN PLANT LOCATION*



The local industry now has only 2 types of mills operating: Non-integrated recycling paper mills and non-integrated pulp abaca mills. Existing paper mills, all recycle-based (non-integrated), account for 10 firms located in Metro Manila (48%) and in the regional provinces (52%).



Most of the paper mills are small by international standards, with capacities lower than 60,000 tons per year. There are only 4 manufacturers capable of producing 70,000 Tons or more per year. As shown in the map above, these are Trust International Paper Corporation, United Pulp and Paper Corporation, Bataan 2020 Inc., and Container Corporation of the Philippines.

LOCATION	COMPANY	PRODUCT LINES	Estimated Effective Capacity, MT/year
Quezon City	Container Corporation of the Philippines	Corrugating medium, test liner, Chipboard, Brown paper	89,000
Samal Bataan	Bataan 2020	Printing/writing, Newsprint, Corrugating. medium, Test liner, Tissue	73,000
Mabacalat, Pampanga	Trust International Paper Corporation	Newsprint, printing and writing	230,000
Calumpit Bulacan	United Pulp and Paper Corporation Inc.	Corrugating medium, test liner	230,000

Locating a new paper plant in San Jose Del Monte (SJDM), Bulacan, is a highly strategic decision that capitalizes on the established "industrial cluster" formed by major players in Central Luzon. By positioning REAM-R-KABLE FIBERS in SJDM, the company benefits from Industrial agglomeration, which creates a shared ecosystem of specialized labor, chemical suppliers, and machine maintenance services that already serve these nearby mills.

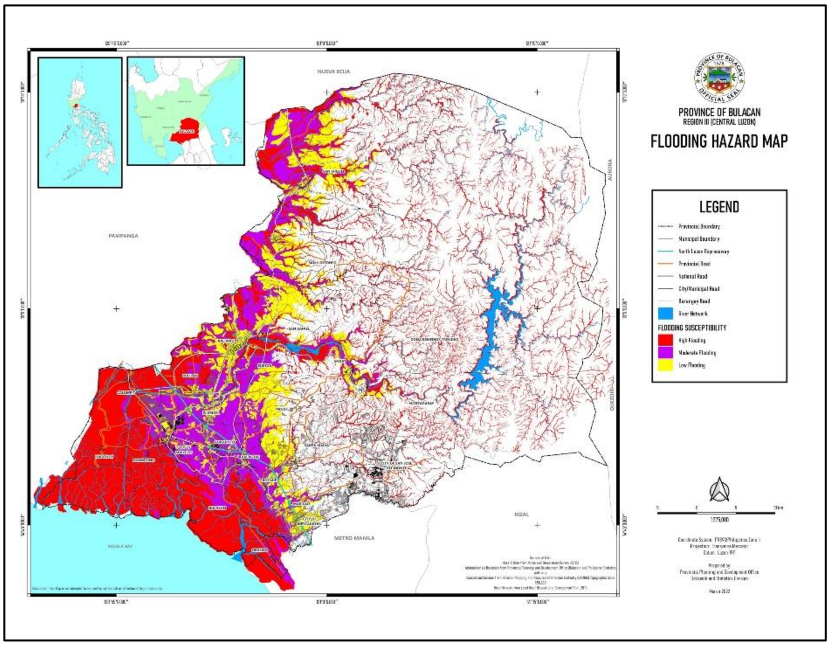
From a logistical standpoint, SJDM serves as a pivotal bridge between the northern production hubs and the massive consumer market of Metro Manila. SJDM offers a unique vantage point that allows a plant to intercept raw material supply chains such as wastepaper collection routes moving toward the established mills while maintaining shorter delivery times to Manila-based clients like the Container Corporation of the Philippines. This combination of shared industrial resources, superior logistics, and geological resilience makes SJDM an ideal strategic site for new paper manufacturing plant.

- *MARKET AREA AND LABOR SOURCES*

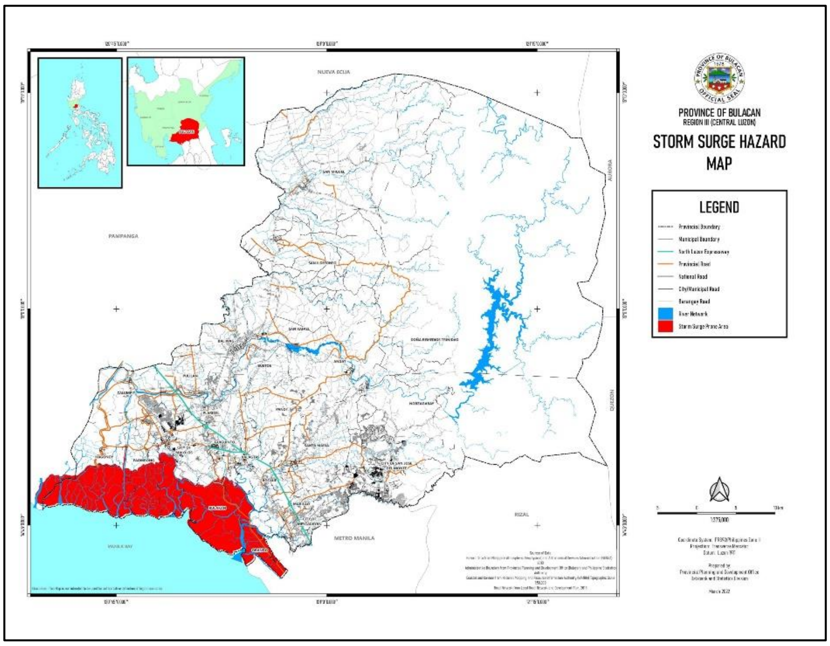
Existing domestic paper and pulp mills located in urbanized areas of Metro Manila and surrounding provinces (Cavite, Laguna, Batangas, **Bulacan**, Pampanga) enjoy the advantages of being near from the sources of high concentration raw material wastepaper. Aside from those mentioned, they also benefit on higher availability of skilled personnel, outsourced services, and proximity to market centers for distribution or marketing of finished products to users.

If compared to mills located in the provinces like in Mindanao, they operate less on wastepaper and more on indigenous fiber sources such as tree plantations, agricultural wastes like rice straw, bagasse and banana, and annual crops like but skilled personnel are costly, plant investment and administration costs are usually higher and freight costs for raw materials and products distribution is high.

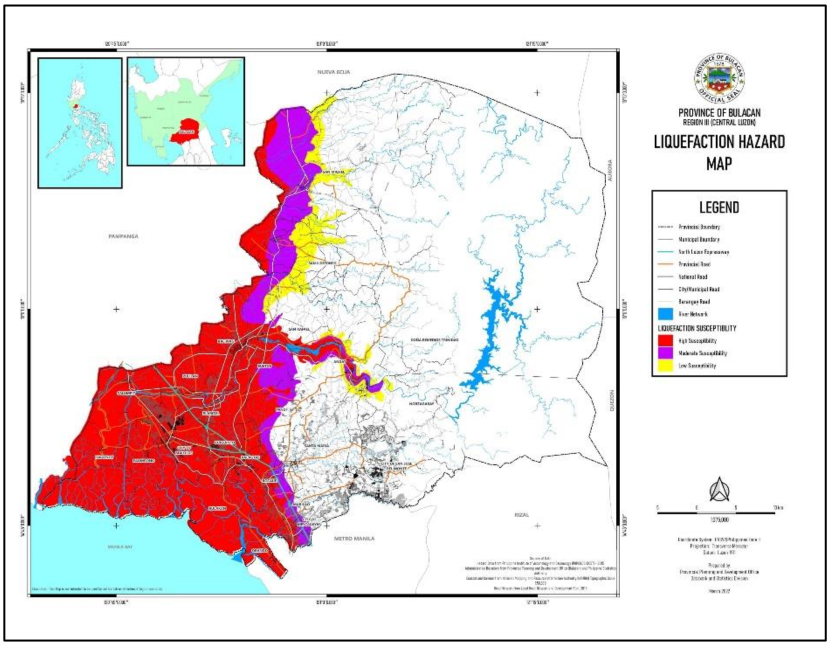
- *CLIMATE CONSIDERATIONS AND OTHER HAZARDS*



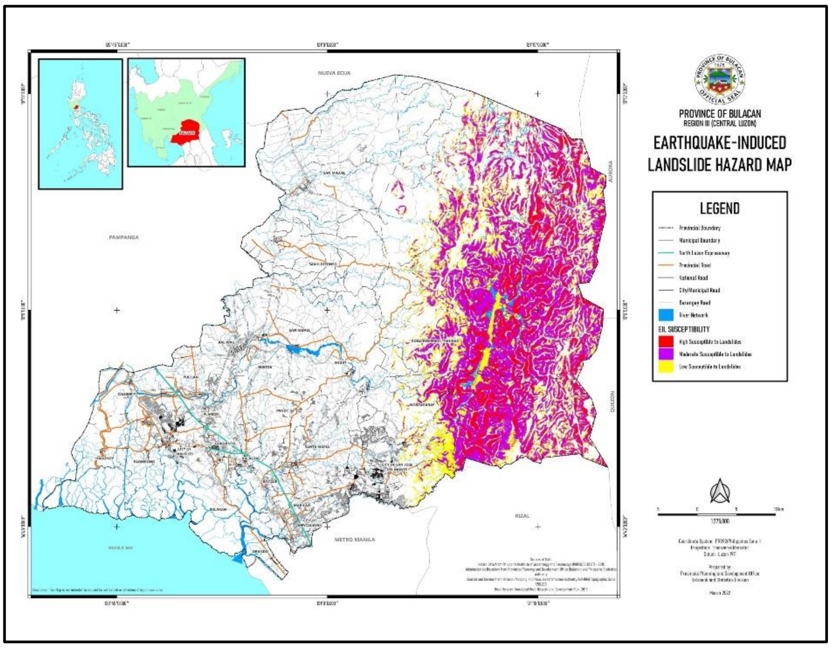
The Flooding Hazard Map for Bulacan Province shows that San Jose del Monte is a suitable site for a paper manufacturing facility, particularly in terms of flood resilience. Based on the color-coded susceptibility key, the highest flooding risks indicated in red for high susceptibility, and purple for moderate susceptibility are found in the low-lying western areas near Manila Bay and along significant river systems. Conversely, San Jose del Monte is located in the eastern part of the province, which is largely free from these hazardous areas. This geographical benefit guarantees that the facility would continue functioning even in severe monsoon seasons or typhoons that generally flood the coastal lowlands



The Storm Surge Hazard Map for Bulacan Province further supports the appropriateness of San Jose del Monte as a safe site for a paper mill. According to the map's legend, areas at risk for storm surge (marked in solid red) are strictly confined to the coastal municipalities adjacent to Manila Bay, including Obando, Meycauayan, Bulakan, and Hagonoy. Due to its considerable inland position and higher altitude compared to the coast, San Jose del Monte is geographically shielded from the severe impacts of sea-level rise and coastal inundation resulting from tropical cyclones. The complete lack of storm surge risk offers an essential level of safety for the substantial industrial investments necessary for paper production



The Liquefaction Hazard Map for Bulacan Province offers compelling geological reasons for situating a paper mill in San Jose del Monte Bulacan. The map highlights zones of significant susceptibility (red) and moderate susceptibility (purple), mainly in the western alluvial plains and coastal areas where soil deposits are prone to being loose and saturated with water. San Jose del Monte, on the other hand, is located entirely in the white zone, signifying no considerable liquefaction risk. This indicates that the base material in this region consists of thicker, more stable soil or bedrock, which is less susceptible to the "liquid-like" behavior of the ground during an earthquake



The Landslide Hazard Map due to Earthquakes for Bulacan Province offers a distinct geographic benefit for setting up a paper mill in San Jose del Monte. The mountainous eastern Sierra Madre area is significantly characterized by high (red) and moderate (purple) susceptibility zones, while the urbanized and industrializing parts of San Jose del Monte predominantly lie within the white and yellow zones, signifying very low to minimal risk. This differentiation is crucial, as it verifies that the chosen location sidesteps the steep, unstable inclines susceptible to mass wasting during earthquakes, guaranteeing that the facility stays accessible and structurally safe

DIFFERENT CONSTRAINTS APPLIED IN DESIGNING THE PLANT

Designing a pulp and paper mill is a massive, capital-intensive engineering challenge. Because these facilities consume vast amounts of raw materials, water, and energy., while producing significant environmental footprints, engineers must balance several competing constraints.

1. **Environmental and Regulatory Constraints**

Environmental compliance is often the strictest constraint, dictating the mill's location, technology, and operating costs.

- **Effluent and Water Management**: Mills require enormous volumes of water. Regulations strictly limit the temperature, pH, biochemical oxygen demand (BOD), chemical oxygen demand (COD), and total suspended solids (TSS) of the discharged water. This forces the design toward Minimum Effluent Mills (MEM) or closed-loop water systems.
- **Air Emissions**: Mills produce odorous sulfur compounds (total reduced sulfur, or TRS) and particulate matter. Designs must incorporate advanced recovery boilers, scrubbers, and electrostatic precipitators to meet air quality standards.
- **Solid Waste**: The process generates large amounts of sludge, ash, and bark. The design must include infrastructure for byproduct utilization (e.g., burning bark for fuel, using lime mud) or secure landfilling.

2**. Raw Material and Fiber Constraints**

The choice of raw material fundamentally shapes the entire mill layout and chemical process.

- **Wood Species Availability**: Whether the mill processes softwoods (like pine, providing long fibers for strength) or hardwoods (like eucalyptus, providing short fibers for smoothness/opacity) dictates the cooking and refining parameters.
- **Alternative Fibers**: If the mill uses recycled fiber or agricultural residues, the design must shift from heavy chemical pulping to intensive de-inking, cleaning, and contaminant-removal systems.
- **Logistics and Storage**: Wood yards must be constrained by seasonal availability. The design must accommodate massive storage areas for logs or chips to prevent rot while ensuring a continuous supply.

3. **Energy and Utility Constraints**

Pulp and paper mills are highly energy-intensive, requiring both massive electrical power and thermal energy (steam).

- **Co-generation (CHP):** Modern mills are designed to be energy self-sufficient or even net exporters of power. The design must balance the steam generated by the recovery boiler or the burning black liquor and the biomass boiler (burning bark) with the electrical demands of the mill via steam turbines.
- **Steam Hierarchy**: The mill must be strictly zoned by steam pressure levels to optimize heat integration across the cooking, bleaching, and paper-drying stages.

4. **Technical and Process Constraints**

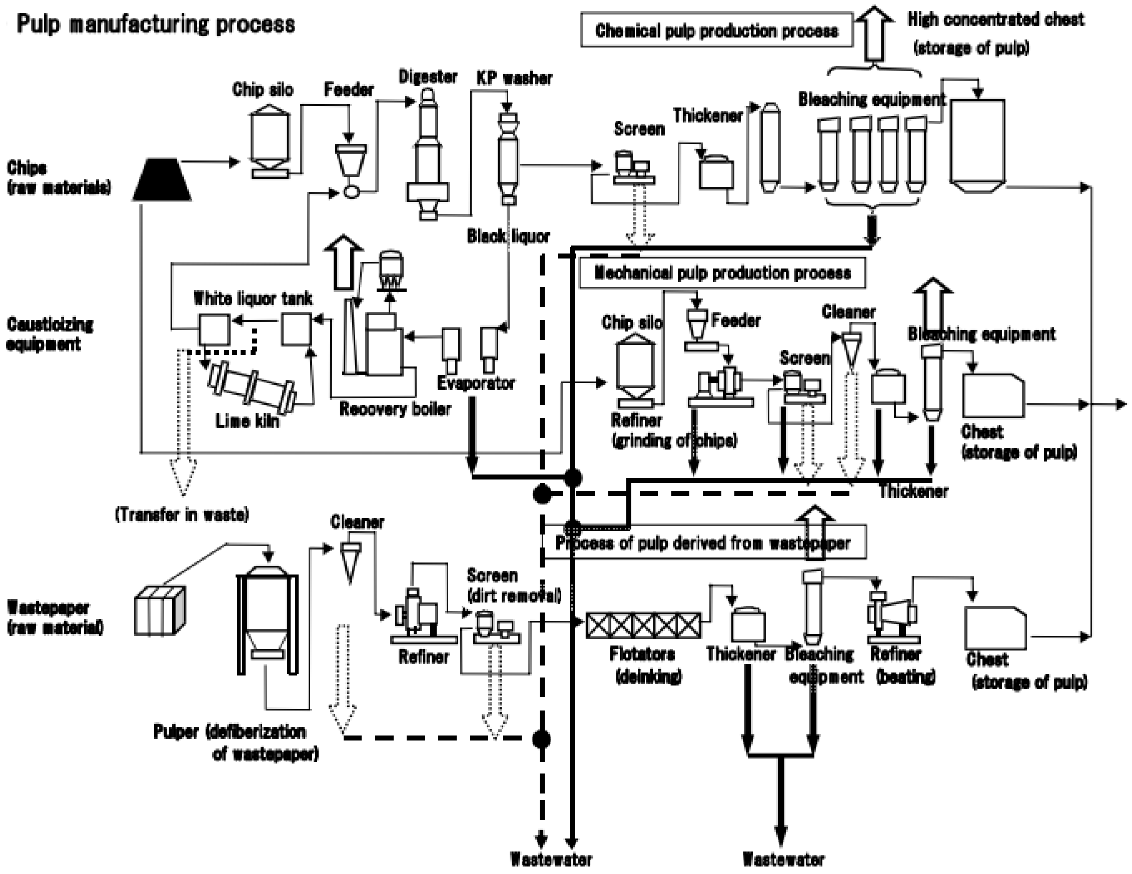
These are the engineering boundaries that ensure the mill actually produces the desired product quality efficiently.

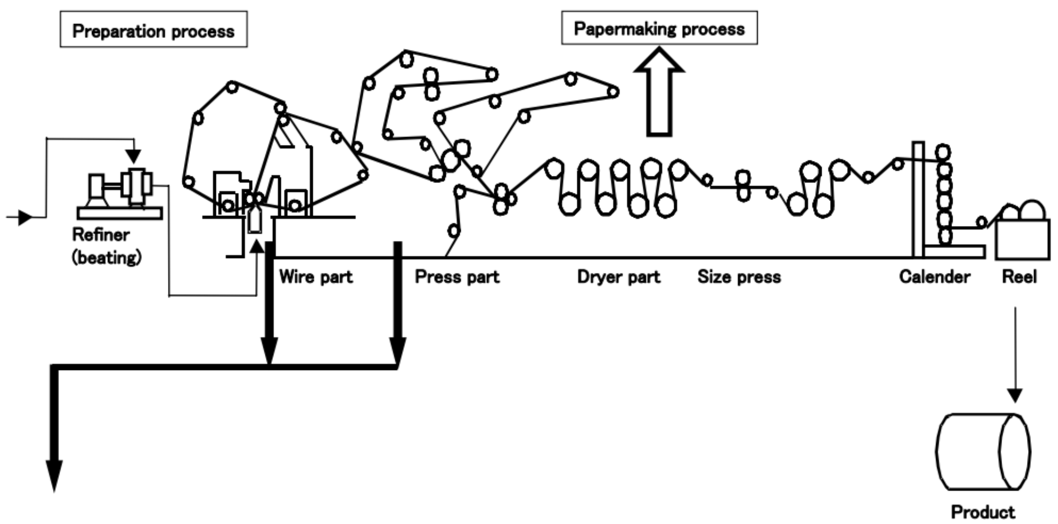
- **Pulping Method**: The design must choose between Chemical (Kraft/Sulfite) pulping (higher strength, lower yield, expensive chemical recovery) and Mechanical/Chemi-mechanical pulping (higher yield, lower strength, massive electricity consumption).
- **Bleaching Sequences:** To avoid toxic chlorinated organic compounds (AOX), designs are legally and technically constrained to Elemental Chlorine-Free (ECF) or Totally Chlorine-Free (TCF) bleaching sequences, utilizing oxygen, ozone, and hydrogen peroxide.
- **Metallurgy and Corrosion**: The chemicals used (sodium hydroxide, sodium sulfide, chlorine dioxide) are highly corrosive. Material selection is a massive cost constraint, requiring extensive use of high-grade duplex stainless steels or titanium in bleaching towers.

5. **Site and Geographical Constraints**

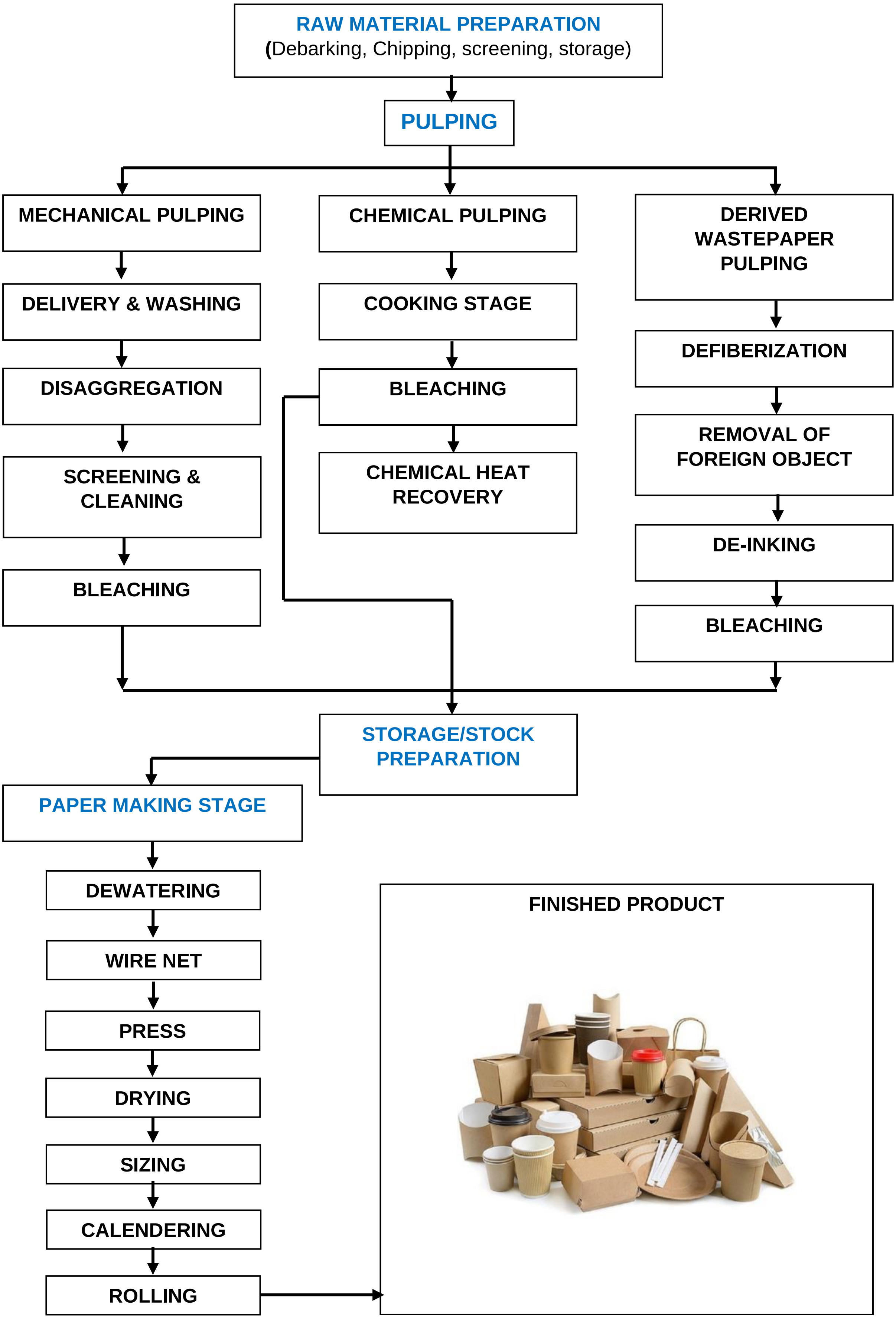
- **Water Source Proximity**: Mills must be near a major river, lake, ocean or dam for water intake and treated effluent discharge.
- **Transportation Infrastructure**: Proximity to rail lines, deep-water ports, or major highways is a hard constraint for moving thousands of tons of raw logs in and finished paper rolls out daily.

PROCESS FLOW DIAGRAM





BLOCK DIAGRAM FOR PAPER AND PULP INDUSTRY



EXPLANATION OF DIFFERENT PROCESSES INVOLVED

RAW MATERIAL PREPARATION:

(1) DEBARKING:

Debarking process is done before the wood can be chipped. The removal of the bark is essential as the bark is often considered a contaminant in the pulping process and can negatively affect the quality of the final product. It is done using mechanical debarking drums or high-pressure water jets.

(2) LOG CUTTING

After debarking, the logs are cut into shorter lengths to fit into the chipper. This process ensures the logs are of a uniform size, making them easier to handle and process.

(3) CHIPPING:

The pre-cut logs are fed into a chipper, where they are sliced into uniform chips. The chipper contains rotating knives that slice the wood into specific sizes. The size and uniformity of the chips are critical for efficient pulping.

(4) SCREENING

After chipping, the wood chips are passed through a screening process where oversized or undersized chips are removed. Only chips of the correct size are sent to the pulping process.

(5) STORAGE AND TRANSPORT

The screened wood chips are stored in large silos or bins until they are needed for pulping. They are then transported to the pulping mill, where they will be processed further.

CHEMICAL PULPING, MECHANICAL PULPING, AND WASTEPAPER PULPING:

METHOD 1: Chemical Pulping

Chemical pulp is produced using kraft pulp processing because this method produces strong pulp, is highly adaptable to any wood species, and pulping chemicals can be recovered and recycled. Moreover, the method converts constituents other than wood fiber, such as lignin, into heat energy using boilers in the chemical recovery process and then makes use of such energy.

(1) COOKING PROCESS

Woodchips and white liquor which is a formed by mixing Sodium Hydroxide (NaOH) and Sodium Sulfide (Na2S) are added into a digester, prepared at elevated temperature and pressure (approximately 160 ℃ and 10 atmospheres) pressures), and divided into cellulose fibers and lignin, among others. The cellulose fibers are consistently rinsed in warm water and directed to the bleaching procedure following the removal of raw materials using a screen.

(2) BLEACHING PROCESS

The fibers are brownish after the cooking process, as they still contain some amount of lignin, etc. Higher brightness can be obtained by removing the remaining lignin with chlorinate bleach chemicals (chlorine, hypochlorite, and chlorine dioxide) and caustic soda in several phases.

(3) CHEMICAL RECOVERY PROCESS

Lignin and hemicellulose excluding fibers, which are separated at the cooking process, and the remaining liquid chemicals are sent to the chemical recovery process. The chemical compound (black liquor) is burned in recovery boilers after being concentrated by vacuum evaporators and is effectively used as heat energy.

METHOD 2: Mechanical Pulping

Mechanical pulping is the process in which wood chips are broken down into fibers using mechanical means only. In Mechanical Pulping, the wood is physically ground in the wood pulping process to separate fibers while retaining most of the lignin.

(1) DELIVERY AND WASHING OF CHIPS

An electromagnetic feeder adjusts the amount of chips that are supplied from a chip silo via a flow conveyor. Next, a measuring conveyer continuously measures their weight. The chips are then sent to the next process after foreign matter such as metallic materials, soil and sand is removed.

(2) DISAGGREGATION PROCESS (REFINER PROCESS)

A refiner is equipped with two rotating disks that face each other. Radial grooves are cut into the surfaces of the disks. Chips are passed between the disks and disaggregated (ground).

(3) SCREENING PROCESS

Screens are used to separate raw materials from foreign objects depending on the size and shape. Cleaners are used to separate raw materials from foreign objects by differences in specific gravity. The foreign objects removed by screens and cleaners are sent to the wastewater treatment process or burned after the dewatering process.

(4) BLEACHING PROCESS

Parts of the screened raw materials are bleached and some are not. Targeted brightness can be obtained by changing the additive rate of bleaching chemicals such as sodium hydrosulfite and hydrogen peroxide.

(5) DRAINAGE PROCESS

The pulp is drained with thickener to increase the concentration, and then continuously stored in the chest. The drained (white) liquor is sent to the wastewater treatment process.

METHOD 3: Derived Wastepaper Pulping

Wastepaper pulp is formed from recovered waste paper or paper trim from printing plants through mechanical dispersion, deinking, and when needed, bleaching process. It is an essential component of resource recycling in modern day paper industry.

(1) DEFIBERIZATION PROCESS

Collected wastepaper and water are put in a pulper to make the treatment concentration about 3 to 5 percent. High-speed mechanical churning for 30 to 60 minutes reverts wastepaper back to the state of pulp. Heating or chemical substances are sometimes applied to facilitate defiberization.

(2) REMOVAL OF FOREIGN OBJECTS

Wastepaper containing such foreign objects cannot be recycled, therefore they are carefully removed with cleaners and screens.

(3) DEINKING PROCESS

Deinking is one way to remove the coloring matter or to clean stained raw materials. Ink is flaked off, dispersed, suspended, and separated from fibers in wastepaper by using mechanical power in addition to alkali and chemicals such as surface-active agents.

(4) BLEACHING PROCESS

In order to further improve the quality of deinked wastepaper, a bleaching process is applied to increase brightness. Wastepaper pulp with the degree of brightness (55 - 75 percent) required for particular application such as newspaper and fine paper is achieved by using bleaching chemicals such as hydrogen peroxide, sodium hydrosulfite and sodium hypochlorite

STORAGE/ STOCK PREPARATION

The following is a description of the beating of raw materials, blending of raw materials, and the addition of fillers, sizes, and chemicals.

(1) BEATING OF RAW MATERIALS

Beating swells fibers increase the flexibility and surface area of fibers, fibrillate, and cut by beating fibers with a disk refiner, etc. The more fibers are beaten, the more their water retention is improved and sheet formation is facilitated.

(2) BLENDING OF RAW MATERIALS

Blending raw materials are to recycle broke generated during the papermaking, coating, and finishing processes and to blend raw materials and recovered white water released outside from preparation and papermaking processes.

(3) ADDITION OF FILLERS, SIZING AGENTS, AND CHEMICALS

Fillers is added to improve opacity, brightness, smoothness, and ink receptivity by filling the void between fibers to improve printability and appearance after printing.

Sizing treatment is designed to prevent ink feathering on the paper for printing or writing, or to add water resistance to wrapping paper and board liners for corrugated fiberboard due to their content and usage conditions.

Chemicals, such as retention aids (that reduce retention loss with fixing flour), strengthening agents for paper (that improve sheet strength of paper by increasing the number of fiber bonding points), slime control agent (that keeps slime from developing by sterilizing microorganisms), antifoaming agent, and dyes, etc., are added as required

PAPER MAKING STAGE

(1) HEADBOX AND WIRE SECTION

The headbox of a paper machine distributes the fiber-water suspension across the entire width of the wire. In this way, a best-possible fiber orientation is achieved. On the wire, the fibers settle next to and on top of one another. At the same time, the water runs off or is sucked away. This forms the sheet.

(2) PRESS SECTION

Further draining process is by mechanical pressure in the press section. Using an absorbent continuous felt cloth the paper web runs through rollers of steel, granite or hard rubber, thus draining it. Pressing compacts the paper structure, its strength is increased and the surface quality significantly improved.

(3) DRYER SECTION

After the press part, the wet paper sheet is heated with the dryer to a moisture content level of 6 - 10 percent.

(4) SIZE PRESS

The size press machine applies starch to paper in the middle of the dryer process in order to give water resistance and surface strength to the paper.

(5) CALENDERING

A calender consists of several steel rollers stacked on top of one another which give the paper web a smooth surface, gloss it, and an even sheet thickness.

(6) REEL AND FINISHED PRODUCT

The paper sent out of the calender is wound up on to a steel shaft (reel) and shipped as a product after finishing.

MANPOWER LISTING

The manpower requirement will be composed mostly of local skilled workers to be hired based on their qualification, experience and good moral character. Priority employment will be given to local residents of San Jose Del Monte Bulacan, and also within the province of Bulacan as well in neighboring areas such as in Metro Manila provided that they possess the necessary qualifications.

SKILL CATEGORIES REQUIRED FOR THE CONSTRUCTION PHASE

CATEGORY	TRADE/SKILL
CIVIL & CONSTRUCTION	Engineers
	Architect
	Safety Officer
	Warehouse man
	Welder
	Pipe Fitter
	Mason/carpenter
	Erector
	Construction Workers
	Tool Keeper
	Operator (heavy equipment)
	Electrician
	Draftsman
	Leadman
	Steelman
	Foreman
	Rigger
	Helper

TOP MANAGEMENT

ROLE	QUANTITY	Job Description
Chief Executive Officer (CEO)	1	Ultimate accountability for plant performance, safety, and profitability.
Chief Operating Officer	1	Manages daily operations across all departments; implements strategic directives.

Chief Technical Officer	1	Oversees process technology, automation, and continuous improvement.
Plant Manager	1	Overall responsibility for production, safety, quality, and cost on site.

SUPPLY CHAIN AND LOGISTICS FOR RAW MATERIALS

Role	Quantity	Job Description
Logistics and Procurement Director	1	Manages all material sourcing, contracts, and inbound logistics
Procurement Manager	1	Executes purchase orders, negotiates with suppliers
Purchasing Officer	2	Process requisitions, track deliveries, manage documentations
Delivery and Receiving Supervisor	1	Oversees weighbridge operations and unloading crews.
Receiving Clerks	2	Sample materials, log incoming quantities, update inventory system.
Forklift Operators	3	Unload trucks, move materials to stockpiles or silos.
Logistic Coordinator	1	Schedules inbound deliveries and coordinates with production.

OPERATING PERSONNEL NEEDED FOR THE PAPER MILL PLANT

Role	Quantity	Job Description
Plant Manager	1	Overall operations, safety, compliance
Operations Manager	1	Oversees daily production targets across pulp and paper lines
Debarker operator	1	Manages raw wood processing
Chipper Operator	1	Processes debarked wood
Heavy Equipment Operator	2	For stockpiling of debarked/chipped woods or log
Recovery Boiler Operator	1	Manages burning of black liquor to recover chemicals and generate steam
Lime Kiln Operator	1	Manages the conversion of green liquor to white liquor
Pulper Operator	1	Manages wastepaper feeding or virgin bale slushing
Chemical Preparation Man	1	
Deinking machine operator	1	Manages wastepaper pulp
Paper Machine Operator	5	Responsible for the papermaking phase from headbox to finishing

Inbound assistant	1	
Outbound assistant	1	
Supervisor	1	

QUALITY CONTROL AND ASSURANCE

Role	Quantity	Job Description
Quality Assurance Manager	1	Implements quality systems; oversees GMP and documentation.
Quality Control Manager	1	Manages laboratory testing and product certifications.
Product Quality Inspectors	1	Conduct periodic sampling and destructive testing.
Document Control Specialist	1	Maintains test records, certificates of conformance.

MAINTENANCE AND ENGINEERING

Role	Quantity	Job Description
Maintenance Manager	1	Overseas the entire department, budgets and contract management
Mechanical Maintenance Engineer	2	Troubleshoots complex mechanical failures
Electrical and Instrumentation Engineer	2	Manages the mill's power distribution, high voltage substations, and control loops
Millwrights	1	Laser alignment of pumps, gearboxes, replacing doctor blades and maintaining hydraulics/pneumatics
Lubrication Technician	1	Daily Tracking and replenishment of automated lubrication systems, manual greasing
Automation/ Control Systems Engineer	1	Specializes in the quality control systems and PLCs

PACKAGING

Role	Quantity	Job Description
Packaging & Logistics Manager	1	Coordinates finished goods packaging and outbound shipping.
Packaging Supervisor	1	Manages strapping machine and boxing/palletizing lines.
Finished Goods Warehouse Supervisor	1	Manages finished goods inventory and FIFO rotation.
Shipping Supervisor	1	Oversees warehouse and dispatch.

Dispatch Clerks	2	Process shipping documents, coordinate truck loading.
Forklift Operators (Finished Goods)	3	Load trucks and organize warehouse storage.

ENVIRONMENT, HEALTH, SAFETY & SECURITY

Role	Quantity	Job Description
Safety Officer	1	Conducts daily safety inspections, risk assessments, fire drills, Lockout/Tagout.
Security Guards	5	Protect property, control access, monitor fuel storage area, prevent theft.

SALES AND MARKETING

Role	Quantity	Job Description
Sales & Marketing Director	1	Drives revenue; manages customer relationships.
Sales Manager	1	Leads sales team; sets targets.
Product Specialists	2	Support architects, contractors, and distributors.
Client Relations Officers	1	Process orders, handle inquiries, manage accounts.
Branding & Advertising Manager	1	Promotes wood grain bricks and tiles; manages digital presence.
Market Analyst	1	Tracks construction trends and competitor pricing.

HUMAN RESOURCE ADMINISTRATION

Role	Quantity	Job Description
HR Director	1	Sets HR policy; ensures labor law compliance.
HR Manager	1	Manages recruitment, training, and employee relations.
Recruitment & Training Officer	1	Hires and trains production staff.
Employee Welfare Officer	1	Handles safety, benefits, and grievances.
Timekeeping & Payroll Specialist	1	Tracks attendance and processes payroll.
General Services Staff	2	Maintains cleanliness of production areas, restrooms, offices, grounds.

FINANCE AND ACCOUNTING

Role	Quantity	Job Description
Finance Director	1	Oversees budget, cost control, and financial reporting.
Chief Accountant	1	Manages accounts payable/receivable, general ledger.
Accounts Payable/Receivable Officer	1	Processes invoices and collections.
Cost Control & Budget Analyst	1	Tracks production costs, energy usage, raw material variances.
Finance Director	1	Oversees budget, cost control, and financial reporting.

MANPOWER TOTAL

DEPARTMENT	NUMBER OF PERSONNEL
Top Management	4
Supply Chain and Logistics for Raw Materials	11
Operating Personnel Needed for The Paper Mill Plant	19
Quality Control and Assurance	4
Maintenance And Engineering	8
Packaging	9
Environment, Health, Safety & Security	6
Sales And Marketing	7
Human Resource Administration	7
Finance And Accounting	5
TOTAL	**80**

LIST OF EQUIPMENT AND SPECIFICATION

RAW MATERIAL HANDLING AND PREPARATION

FOR INPUT WOOD/LOG

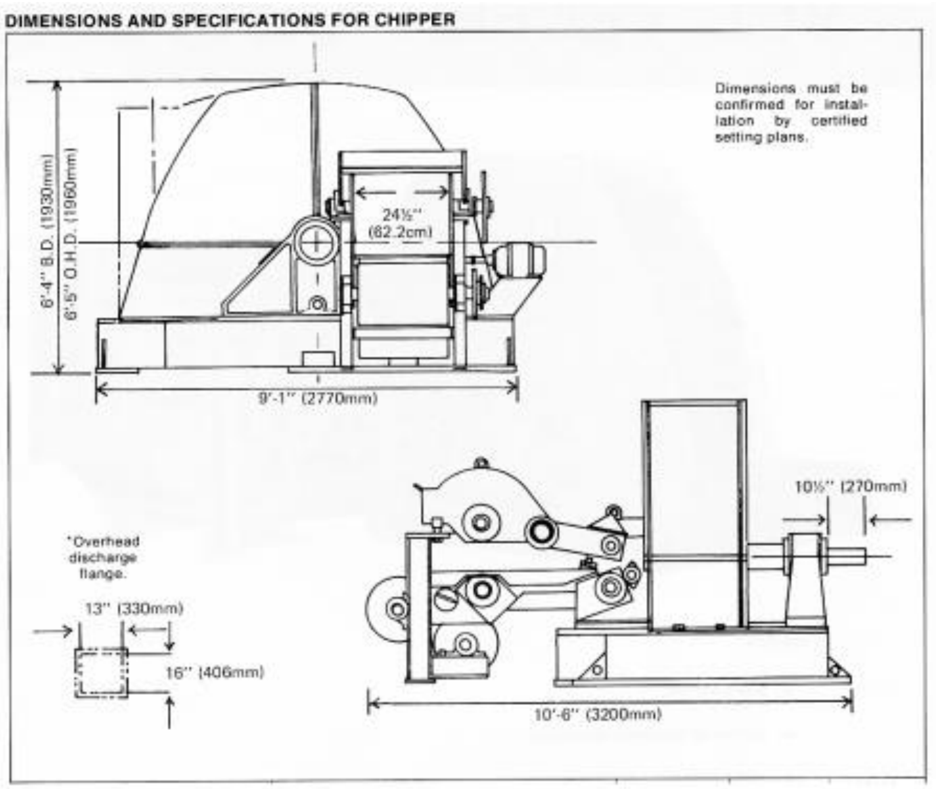
EQUIPMENT	DESCRIPTION/ USE
 ROLLER WOOD DEBARKER	Roller debarker is used for removing the log wood bark. It has wide applications for stripping the branches, irregular woods, wood sections, etc. After tree bark removal, you can use the wood chipper to continue the further procedure.

SPECIFICATIONS:

Model	SL-S500	SL-S600	SL-S700	SL-S800
Wood diameter (mm)	50-200	50-300	50-400	50-500
Wood length(m)	2-4	2-5	2-5	2-6
Capacity(T/H)	5-6	6-8	10-12	15-18
Peeling rate	＞95%	＞95%	＞95%	＞95%
Roller diameter(mm)	250	274	370	540
Roller rotational speed(rpm)	230	230	200	180
Roller length(mm)	580	580	580	580
Power(kW)	7.5*2	7.5*2	11*2	15*2
Weight(T)	2.5	2.8	3.5	6
Size(L*W*H)(m)	6*2.1*1.5	6.02*2.26*1.52	6*2.5*1.6	6*2.8*1.7

EQUIPMENT	**DESCRIPTION/ USE**
 WOOD CHIPPER	Essential in producing uniform wood chips from whole logs and/or waste wood. The equipment helps in maximizing the use of excess material and increase the yield of usable fiber.

SPECIFICATIONS:



NO. OF KNIVES	SPOUT SIZE	INFEED BELT SPEED (FPM)			CAPACITY UNITS/ HOUR	DRIVE MOTOR H.P.	MAX. DISC R.P.M.
		¾" CHIPS	1" CHIPS	1½" CHIPS			
6	7" x 24½"	189	252	378	25	150-200	514
8	7" x 24½"	249	331	Not Available	25	150-200	514

Feedworks: Drive – 10 H.P., Motoreducer; Speedroll – 3 H.P., 900 R.P.M. motor.

EQUIPMENT	DESCRIPTION/ USE
 ROUND SILO CHIP STORAGE	Silo diameters can be up to 25 m for bark, and up to 42 m for chips. Silos can be made of steel or concrete, with or without insulation depending upon the climate. Silo storage helps maintain constant chip moistures.

FOR INPUT RECYCLED WASTEPAPER HANDLING

EQUIPMENT	DESCRIPTION/ USE
 BALE BREAKER	Efficiently breaks apart tightly compressed waste paper bales, preparing raw material for the pulping process. It reduces pulping time, protects downstream equipment from damage, and improves overall mill productivity.

SPECIFICATIONS:
-100-900 TPD (tons per day capacity)
-Dual Motor Drive system
-rotating drum mechanism

PULPING EQUIPMENT

EQUIPMENT	DESCRIPTION/ USE
 PM20 DOUBLE DISC REFINER	Pulp refiner is a mechanical device used to refine pulp by breaking down fibers and improving their properties. It consists of two or more rotating discs, typically made of steel, that create a shearing action on the pulp.

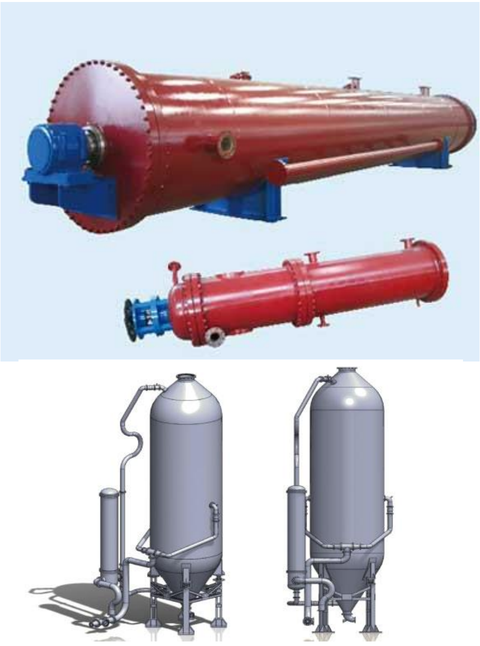
SPECIFICATIONS:

MODEL	PM20	PM26	PM34	PM42
Disc Diameter (mm)	20-24	26-30	34-38	42-48
Min. Flow (L/min)	400	800	1500	2500
Max. Flow (L/min)	2500	5000	10000	15000
Max. Motor Power (KW)	315	500	900	1800

EQUIPMENT	DESCRIPTION/ USE
 HYDRAPULPER	Bales of waste paper are loaded into a circular tank containing water called a hydra pulper. Hydra pulper has a very powerful agitator at the bottom which breaks up the bales into small pieces. The hydra pulper is used mainly for crush pulp board, waste paper, waste carton, and deinking, pulp purification.

SPECIFICATIONS:

Volume	1	2	3	5	102
Capacity (t/d)	2-5	6-10	10-15	15-25	30-40
Concentration (%)	2-15				
Power (kw)	7.5	15	22	55	110

EQUIPMENT	DESCRIPTION/ USE
 DISPLACEMENT PULP DIGESTER	Digesters are the key piece of chemical pulping equipment. They house wood chips inside and processed them using caustic chemicals at rapid pressure changes and heat. A pulp digester is an equipment that is used to deal with plant fiber raw materials, and it is a specialized equipment used in the chemical pulping process. This chemical cooking process determines the final quality and properties of paper pulp. It finally has an influence on the quality and strength of the product.

SPECIFICATIONS:

Model	ZJG3	ZJG4	ZJG5	ZJG6	ZJG7	ZJG8	ZJG9
Effective volume (m^3)	110	135	175	225	250	330	400
Diameter (mm)	3600	4000	4500	4500	4500	5600	5800
Design Pressure (Mpa)	0.9/1.2						
Design Temperature	180/240						

EQUIPMENT	DESCRIPTION/ USE
 BLOW TANK	Large cylindrical vessel which functions as intermediate storage of the cooked pulp from the digester, and from which the pulp is discharged in an even flow to a washing process.
SPECIFICATIONS: -SIZE- 80m^3,150m^3,225m^3, 330m^3, 500m^3, 1050m^3 s -Main materials- include carbon steel, 316L steel, stainless steel and composite plate, duplex stainless-steel structure.	

STOCK PREPARATION EQUIPMENT (WASHING, SCREENING & CLEANING)

EQUIPMENT	DESCRIPTION/ USE
 REFINER	Double disc refiner is pulp refining equipment traditionally installed in stock preparation system for both waste paper and virgin pulp, to improve fibers characteristics in order to get a better sheet formation and enhance its properties. It is available for continuous beating of chemical wood pulp, mechanical pulp and waste paper pulp.

SPECIFICATIONS:

Type	DD-500	DD-550	DD-600	DD-660
Dia.of plate (mm)	Φ500	Φ550	Φ600	Φ660
Capacity (t/d)	20-60	30-80	40-100	50-150
Power (kw)	132-200	160-250	185-315	280-355
Concentration (%)	3-5	3-5	3-5	3-5

EQUIPMENT	DESCRIPTION/ USE
 HIGH SPEED WASHER	High speed washer is a pulp washing and concentrating equipment in stock preparation system for waste paper pulp. It is able to wash and thicken waste paper pulp efficiently through the centrifugal force generated by the high-speed rotation of rubber covered roller, as well as the extrusion force between the roller and the net.

SPECIFICATIONS:

Type	ZGX-1000	ZGX-1500	ZGX-2000	ZGX-2500	ZGX-3000
Inlet consistency	0.5-1.5 %				
Outlet Consistency	10-17 %				
Capacity (T/D)	20-40	30-50	60-80	80-100	100-120
Wire Width (mm)	1152	1690	2072	1360*2	1560*2
Power (KW)	11	18.5	22	30	37

EQUIPMENT	DESCRIPTION/ USE
 DISK FILTER	It is the best choice for white water recovery and pulp concentration. It can save energy and water, reduce fiber loss and prevent pollution.

SPECIFICATIONS:

MODEL	ZNP 25	ZNP 36	ZNP 45	ZNP 52
FOR WHITE WATER RECOVERY				
Filter disc diameter (mm)	Φ2500	Φ3600	Φ4500	Φ5200
Filter disc quantity (n)	4-12	6-16	8-20	10-20
Filter area (m^2)	7.5	15	26	32
Main drive motor power kw	5.5-11	7.5-15	11-18.5	22-37
Concentration of inlet white water (%)	0.35-0.45			
Outlet concentration (%)	3-5			
White water throughput (m^3/m^2h)	1.6-4.0			
FOR PULP THICKENING				
Filter disc diameter (mm)	Φ2500	Φ3600	Φ4500	Φ5200
Number of filter disc quantity (n)	4-12	6-16	8-20	8-20
Filter area per Disc (m^2)	7.5	15	26	32
Main drive motor power (kw)	5.5-11	7.5-15	11-18.5	22-37
Pulp type	chemical wood pulp, deinked pulp, straw pulp			
Inlet consistency (%)	0.6-1.2			
Outlet consistency (%)	8-12			
Production capacity ($t/m^2 \cdot d$)	0.5-1.2			

EQUIPMENT	DESCRIPTION/ USE
 VACUUM DRUM WASHER	Vacuum drum washer is a common pulp washing equipment for bleaching workshop and extraction of pulp like wood pulp, straw pulp and other types of pulp

SPECIFICATIONS:

CRITERIA	UNIT SPECIFICATION
Model	ZXIV20
Filter Area	$20m^2$
Feeding Pulp Concentration	1.0-2.0
Outlet Pulp Concentration	8-12
Drum Specification	Φ3000×2300㎜
Capacity	106adt/d dissolving pulp(bleached)
Power	11（variable-frequency ac motor ）
Broken Helix	Φ500×2300
Screw motor power	5.5 kw
Spiral Speed	88 rpm
Material Clamp Device Specifications	Φ500×2800
Material Clamping device power	5.5 kw
Material Clamping device speed	60 rpm
Stirrer Device specification	Φ500×2800
Stirrer Device power	7.5 kw
Stirrer Device Speed	88 rpm
Fan Delivery	$2000m^3$/h
Fan discharge pressure	1960 pa
Power for motor of fan	2.2 kw

FLOATATION DEINKING

Flotation deinking cell is suitable for floatation deinking all kinds of waste paper pulp. Some light weight plastic can also be removed.

The air will be used in pulp time after time so the ink removing and air using rate are improved effectively.

SPECIFICATIONS:

TYPE	THROUGHPUT $(m^3/h$	Cell Diameter (mm)	Output (T/D)	Concentration (%)	Ink Removing Rate	Pulp Inlet pressure
MAC-s2500	40-60	2500	15	0.8-1.5	≥95	0.25-0.3
MAC-3000	110-160	3000	40	0.8-1.5	≥95	0.25-0.3
MAC-4000	140-210	4000	50	0.8-1.5	≥95	0.25-0.3
MAC-5000	270-420	5000	100	0.8-1.5	≥95	0.25-0.3
MAC-6000	420-625	6000	150	0.8-1.5	≥95	0.25-0.3

EQUIPMENT	DESCRIPTION/ USE
 REJECT SEPARATOR	Reject separator is mainly used for fiber relief and impurities separate for tailing in the coarse screen system for waste paper pulp. It has the benefits of less fiber loss and low production cost.

SPECIFICATIONS:

Model	Standard (mm)	Inlet consistency (%)	Reject consistency (%)	Capacity (T/d)	Power (kw)	(L×W×H): (mm)
LZSZ1	Φ280	1~3.5	15~20	10~20	37KW	2610×1925×1010
LZSZ2	Φ380	1~3.5	15~20	20-35	55KW	3200×2020×1195

EQUIPMENT	DESCRIPTION/ USE
 FIBER SEPARATOR	Fiber separator is pulp defibering equipment in stock preparation system for waste paper pulp. It has stronger removal capacity for the impurities (staples, gravel, plastic film, plastic paraffin and asphalt) in the waste papers, making the pulp clean and thus improving the product quality.

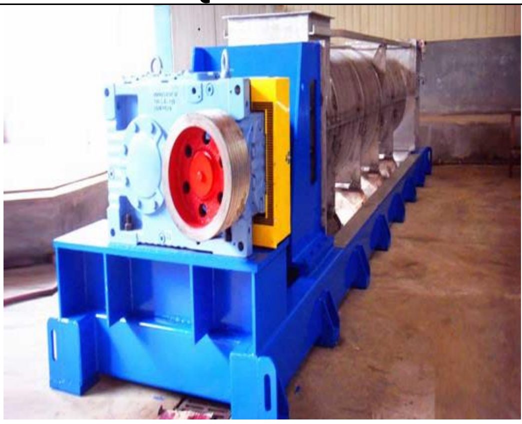
SPECIFICATIONS:

Type	ZDF2A	ZDF3B	ZDF4B	ZDF5B	ZDF6B
Impeller diameter (mm)	560	720	980	1100	1320
Capacity (t/d)	30-50	60-80	100-180	160-260	240-360
Feeding Pulp Concentration	2.5 – 4.0				
Power (kw)	37	55	90	132	185

EQUIPMENT	DESCRIPTION/ USE
 TWIN ROLL PRESS	The twin-roll press is highly efficient equipment for automatic washing and extraction of filtrates and pulp concentrates. It is suitable for black liquor extraction and concentration of pulp, such as non-wood pulp, bamboo pulp, bagasse pulp, wood pulp, and waste paper pulp.

SPECIFICATIONS:

MODEL/ITEM	Drum length (mm)	Drum dia. (mm)	Drum speed (rpm)	Feed (%)	Discharge (%)	Production Capacity		
						Straw/ bagasse pulp	Bamboo pulp	Wood pulp
AMSG712	1200	700	2~20	3.5~10	20~35	30~40	60~80	80~100
AMSG915	1550	900	2~15	3.5~10	20~35	40~60	85~120	130~160
AMSG920	2000	900	2~15	3.5~10	20~35	55~85	125~160	175~210
AMSG928	2820	900	2~15	3.5~10	20~35	80~120	165~220	250~300
AMSG935	3500	900	2~15	3.5~10	20~35	110~150	225~280	310~370
AMSG940	4000	900	2~15	3.5~10	20~35	120~170	220~310	350~400
AMSG945	4500	900	2~15	3.5~10	20~35	160~180	285~350	400~480
AMSG1532	3200	1500	2~15	3.5~10	20~35	220~350	450~600	700~800
AMSG1540	4000	1500	2~15	3.5~10	20~35	400~550	800~1000	1200~1500
AMSG1550	5000	1500	2~15	3.5~10	20~35	500~650	1000~1300	1700~1800
AMSG1572	7200	1500	2~15	3.5~10	20~35	750~1000	1500~1800	2500~3000

EQUIPMENT	DESCRIPTION/ USE
 SINGLE SCREW PRESS	Single screw pulp press machine is used for pulp dewatering, black liquor extraction of wood pulp, bamboo pulp and wheat straw pulp after cooked and high-concentration of recycled pulp waste.

SPECIFICATIONS:

Model/Specification	AMDL-I 600	AMDL-I 750	AMDL-1000	AMDL- 1500
Screw Diameter (mm.)	600	750	1000	1500
Production (Kt/d)	50-100	100-120	200-350	500-750
Power (Kw)	90	160	315	500
Net Weight (Kg)	9200	12500	19000	38000
Inlet Consistency (%)	4~12	4~12	4~12	4~12
Outlet Consistency (%)	25~30	25~30	25~30	25~30

 PULP GRAVITY DISC THICKENER	Pulp gravity disc thickener is mainly used for thickening of low consistency pulp. Different from traditional vacuum disc filter, it doesn't need necessary installation height of water log. It can replace traditional pulp disc gravity thickener.

SPECIFICATION:

ITEM	ZNP 26	ZNP 35
Diameter (mm)	2600	3500
Per-dis filtration area (m^2	7.5	15
Capacity	0.8—3t/m2.d 0.3-2	
Inflow Consistency (%)	0.3-2	0.3-2
Outflow Consistency (%)	3~5	3~5
Washing pressure for Sieve cleaning (mpa)	0.6-0.7	

VIBRATING SCREEN

vibrating screen is mainly used for screening and dehydration of primary pulp, short fiber pulp, and special pulp to improve the quality and frequent production efficiency of paper pulp. It makes paper pulp separate the impurities and water by vibrating method

SPECIFICATIONS:

TYPE	a	b	c
Screening Area (m^2	0.5	0.9	1.8
Sieve Plate Size (mm)	1000*500	1800*500	1800*1000
Concentration (%)	1-1.5		
Capacity (t/d)	3-5	12-30	25-75
Power (kw)	2.2	3	4

EQUIPMENT	DESCRIPTION/ USE
 CENTRIFUGAL SCREEN	Centrifugal screen is used for the screening and dehydration treatment of pulp. It mainly deals with mechanical pulp and high-concentration pulp and is used in the independent filtering section.

SPECIFICATIONS:

Model	OPS-082	OPS-122	OPS-202	OPS-312	OPS-382	OPS-482	OPS-622	OPS-802
Throughput m^3/min	6-10	9-15	15-25	24-39	29-47	36-49	47-77	61-100
Work Consistency	<1.2	<1.2	<1.2	<1.2	<1.2	<1.2	<1.2	<1.2
Inflow part diameter (mm)	250	300	400	500	500	600	600	800
Reject Diameter (mm)	80	80	100	125	150	150	200	200
Machine Height (mm)	1636	2042	2392	2279	3120	3275	3622	3930
Machine length (mm)	1862	2804	2804	3580	4062	4396	4502	5275
Machine Width (mm)	977	1499	1499	2057	2351	2524	2642	3170
Motor Power Input (kw)	18.5	37	55	75	90	132	160	250
Machine Weight (kg)	1445	2833	4360	6679	8177	10818	13960	18200

 PULP CHEST AGITATOR	Pulp chest agitator is used to stir pulp circulatory in square pulp pool. By using the equipment, the pulp liquid is maintained in suspension condition, so it ensures that the uniform distribution of paper components in pulp.

SPECIFICATIONS:

Type	φ500	φ700	φ750	φ1000	φ1250
Slurry tank volume (m3)	20-30	40-60	50-80	70-100	90-120
Slurry Tank Concentration (%)	≤5				
Motor power (Kw)	7.5	11	15	22	30

EQUIPMENT	DESCRIPTION/ USE
 FAN PUMP	Fan pump propels the dilute slurry of papermaking furnish towards the headbox of a paper machine.

SPECIFICATIONS:

Model	Flow		Head	Speed	Power	
	m3/h	L/s	M	Rpm	MODEL	KW
150SJ100-9	100	27.8	9	1450	Y112M-4	4
200SJ150-12	150	41.6	12	1450	Y132M-4	7.5
200SJ150-7	150	41.6	7	1450	Y132S-4	5.5
200SJ180-6	180	48.5	6	1450	Y132S-4	7.5
250SJ250-10	250	69.4	10	1450	Y160L-4	11
250SJ330-6	330	91.6	6	970	Y160M-6	7.5
250SJ500-9	500	138.9	9	1450	Y180M-4	18.5
250SJ500-14	500	138.9	14	1450	Y200L-4	30
250SJ500-30	500	138.9	30	1450	Y250M-4	55
300SJ600-4	600	166.7	4	970	Y160L-6	11
300SJ800-12	800	222.2	12	1450	Y225S-4	37

300SJ800-30	800	222.2	30	1450	Y280M-4	90
300SJ1000-5	1000	277.7	5	970	Y200L-6	2
300SJ1000-9	1000	277.7	9	970	Y250M-6	37

PUMP

Pulp pump is used for conveying the pulp.

Model	Flow (m3/h)	Head (m)	Speed (r/min)	efficiency η (%)	Power (KW)	Suction lift (m)	Concentration (%)
80LXL-Z-6	32-50	7-5	1450	60	2.2	2.5	4-6
80LXL-Z-12	35-60	14-9	1450	60	4	2.5	4-6
80LXL-Z-15	40-65	16-12	1450	60	5.5	2.5	4-6
80LXL-Z-20	35-65	22-18	1450	60	5.5	6	4-6
80LXL-Z-25	40-75	26.5-22	1450	60	7.5	6	4-6
80LXL-Z-30	55-75	32-21	1450	60	11	6	4-6
80LXL-Z-35	56-76	36-25	1450	60	15	6	4-6
80LXL-Z-40	60-80	41-32	1450	60	15	6	4-6
100LXL-Z-18	53-134	18.3-14	1450	60	7.5	7	4-6

BLEACHING EQUIPMENT

EQUIPMENT	DESCRIPTION/ USE
 BLEACHING TOWER	During the pulp production process, the bleaching tower is used to remove lignin and pigments from the wood pulp, thereby making the pulp whiter. High-Intensity Bleaching Tower is suitable for bleaching 25% to 32% concentration of the pulp. It could efficiently increase the whiteness of the paper and reduce bleaching costs.

SPECIFICATIONS:

Bleaching Capacity	Based on the actual production capacity
Pulp Temperature	60-85
Bleaching Concentration (%)	25-32
Sizing consideration (%)	4-5
Bleaching time	60-90
Blade Impeller diameter	φ900-φ1000
Equipped with a blender	BTA800-JBTA1000

 CHEMICAL MIXER	Medium consistency mixer is used to mix the chemicals and pulp in the pulp bleaching process. The good mixing efficiency can provide homogeneous reaction for bleaching process, save the chemicals, and reduce the energy consumption,

SPECIFICATIONS:

Pulp Flow	15L/s, 30L/2, 55L/s
Productivity	100t/d, 200t/d, 350t/d
Rotation Rate	1500r/min
Pulp Consistency (%)	8-12%
Motor Power	11kw, 30kw, 45kw
Material	Cast steel, stainless steel, titanium steel

PAPER MAKING STAGE

EQUIPMENT	DESCRIPTION/ USE
 PRESSURIZED HEADBOX	Headbox is to distribute the papermaking stock uniformly across the width of the wire section. The papermaking stock pumped in a pipe is converted to a uniform rectangular flow with absolutely the same flow rate and flow direction across the wire width.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	20-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 1000 m/min
Capacity	Up to 1000 tpd

EQUIPMENT	DESCRIPTION/ USE
 WIRE SECTION	Wire section performs the fundamental transformation from fiber suspension to coherent paper sheet, making it the most critical section influencing final paper properties.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	20-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 1000 m/min
Capacity	Up to 1000 tpd

EQUIPMENT	DESCRIPTION/ USE
 PRESS SECTION	press section is engineered to deliver maximum dewatering efficiency, improved sheet strength, and stable paper web transfer for all types of paper and board machines. Designed with advanced press roll configurations and optimized loading systems, it ensures lower steam consumption, reduced energy cost, and enhanced paper quality.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	20-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 1000 m/min
Capacity	Up to 1000 tpd

EQUIPMENT	DESCRIPTION/ USE
 DRYER SECTION	Dryer cylinder is designed for high working pressure, its medium to heat is vapor. The dryer cylinder will have a great impact on the final moisture content of the paper and will also consist of the greatest energy usage in the paper making process.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	20-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 1000 m/min
Capacity	Up to 1000 tpd

EQUIPMENT	DESCRIPTION/ USE
 SIZE PRESS	Size Press is used for applying starches, pigments, or chemicals to the surface of the product. The chemicals applied can be used to strengthen the product or to improve the printing quality of the product or even to improve the softness of the product.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	20-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 1000 m/min
Capacity	Up to 1000 tpd

EQUIPMENT	DESCRIPTION/ USE
 CALENDAR	Calendar consists of several rollers arranged vertically one upon the other, by running the almost dry paper web between the rollers under high pressure, the paper is compacted and smooth. The pressure applied to the web by the rollers determines the finish of the paper. To sum up, calendars increase the quality of the final product, make the paper surface extra smooth and glossy, gives it a more uniform thickness to the final product.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	13-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 2200 m/min
Capacity	Up to 1000 tpd

EQUIPMENT	DESCRIPTION/ USE
 REWINDER MACHINE	Rewinder is used to process these jumbos made from paper making machine into manageable and sellable products, the width and tightness of rewinding meets the requirements of finished sheet rolls. In the process of rewinding, it can also remove the bad-quality paper sheets, adhesive end breakage.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	13-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 2200 m/min
Capacity	Up to 1000 tpd

 POPE REEL	**Reel** is a key component of the paper machine finishing section, designed to wind the paper web into uniform jumbo rolls with stable tension control and excellent winding quality. Built for high-speed, continuous production, the pope reel ensures smooth operation from press section to rewinder, delivering consistent roll density and superior paper handling performance.

SPECIFICATIONS:

Model	1575-6600
Paper Grade	All Kinds of paper
GSM range	13-500 g/m2
Trim Width	Up to 6600
Working Speed	Up to 2200 m/min
Capacity	Up to 1000 tpd

CHEMICAL RECOVERY EQUIPMENT

- Recovery boiler
- Evaporators
- Causticizing plant
- Lime kiln

UTILITIES & SUPPORT EQUIPMENT

- Boilers and steam turbines
- Water treatment plants
- Effluent treatment plants
- Air compressors
- Pumps and pipelines

LIST OF MATERIALS AND ACCESSORIES

The paper and pulp industry rely heavily on a wide range of raw materials that play a crucial role in determining the quality, texture, and sustainability of the final product. Key materials in paper production include wood pulp, non-wood fibers and other cellulosic material, recycled paper pulp, and various chemicals. The following provides an in-depth look at these inputs and their importance.

INDUSTRY INPUTS/ RAW MATERIALS

A. VIRGIN/ WOOD PULP

Woods are considered primary raw material in paper industry. It is sourced from both hardwood and softwood trees:

- **Softwood Pulp**: Derived from coniferous trees like pine and spruce, this pulp contains long fibers that enhance the strength and durability of paper, making it ideal for products like printing and writing paper.
- **Hardwood Pulp**: Sourced from trees such as eucalyptus and birch. It has shorter fibers that increase paper bulk but generally result in a weaker end product.

B. NON-WOOD PULP

Non-wood fibers encompass a wide range of plant materials that serve as sustainable alternatives to wood pulp. Common sources include abaca, banana, rice straw, wheat straw, bagasse, and bamboo. Furthermore, recent innovations have enabled the use of agricultural residues, such as waste from pineapple plantations in paper production.

These fibers often need specific processing methods and can impart distinct characteristics to the paper. Non-wood pulps generally have lower tensile strength than wood pulps but may offer advantages in terms of cost-effectiveness and environmental sustainability.

Abaca pulp: The Philippines is the world's primary source of abaca fiber, used for cordage and pulp for specialty paper production. It is a non-wood pulp derived from the fibers of the abaca plant (Musa textilis). Known for its exceptional strength and durability, abaca pulp is highly valued in a variety of applications

C. Recycled Wastepaper

Recycled paper pulp is produced by processing used paper products through a process involving deinking and breaking down the fibers or fiber separation. The utilization of recycled materials offers significant environmental advantages, such as waste reduction, conservation of natural resources, and decreased energy consumption and greenhouse gas emissions associated with paper manufacturing. Due to the lack of virgin wood pulp, the Philippine paper industry primarily relies on recycled wastepaper as its main raw material.

CHEMICALS AND ADDITIVES USED IN PULP AND PAPER INDUSTRY

Pulping Chemicals

- **Caustic Soda (Sodium Hydroxide):** Used to break down wood chips into pulp. It is an indispensable chemical in the pulp and paper industry, with applications ranging from pulping and bleaching to wastewater treatment and chemical recovery
- **Sodium Sulfide**: Acts as a cooking liquor in the Kraft process, helping to dissolve lignin.


Bleaching Agents

- **Chlorine Dioxide (ClO2):** Commonly used for bleaching pulp to achieve a high level of whiteness.
- **Hydrogen Peroxide (H2O2):** An environmentally friendly bleaching agent that helps brighten pulp.
- **Sodium Hydrosulfite –** is an effective in bleaching recycled pulp. Sodium hydrosulfite is a strong reducing agent and can effectively react with the dyes in the pulp.



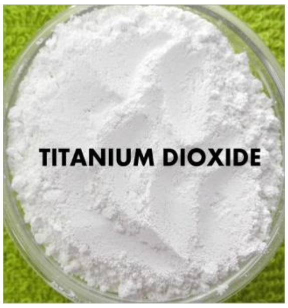
Sizing Agents

- **Alkyl Ketene Dimer (AKD):** A popular sizing agent that provides water resistance to paper.
- **Alkenyl Succinic Anhydride (ASA):** Another sizing agent used for enhancing the paper's resistance to water.

Fillers

- **Calcium Carbonate (CaCO3)**: Used to improve brightness and opacity while reducing costs.
- **Clay (Kaolin)**: A common filler that enhances smoothness and printability of paper.
- **Titanium Dioxide (TiO2):** Added for its brightness and opacity properties.





Binders and Retention Agents

- **Polyethyleneimine:** A retention agent that helps bind fillers to fibers, enhancing dewatering during production.

Wet and Dry Strength Additives

- **Epichlorohydrin:** Provides wet strength to paper, ensuring it retains integrity when wet.
- **Cationic Starch:** Improves both wet and dry strength by enhancing fiber bonding.

Optical Brightening Agents- These agents are added to enhance the brightness of paper by increasing its fluorescence under UV light.

Surfactants- Used in deinking processes during recycling, surfactants help remove ink from recycled fibers.

Other Chemicals

- **Lime (Calcium Oxide):** Used in various processes including causticizing and ph adjustment.

DESIGN COMPUTATION OF:

a. ALL EQUIPMENT:

DEBARKING EQUIPMENT

WOOD QUALITY	WOOD SPECIES	REQUIRED DEBARKING TIME IN MINUTES
Trees easy to debark, for pulp wood	Oak, maple, pine, tropical hardwood species	10-15
Normal to debark, soft wood for pulping	Pine, spruce	20-25
Normal to debark, hard wood for pulping	Birch	30
Fresh groundwood	Spruce	30-40
Dry hardwood for pulping	Aspen, birch	40-60
Dry groundwood	Aspen, birch	60

PAPER MAKING EQUIPMENTS

M/C PRODUCTION CALCULATION

Let:

M/C Speed = 200 mt/min

Deckle = 3.2 mt

GSM = 50

$$\boldsymbol{M}/C\ PROD = \frac{SPEED\ \times DECKLE\ \times GSM\ \times 60}{10^6}$$

$$M/C\ PROD = \frac{200 \times 3.2\ \times\ 50\ \times\ 60}{10^6}$$

$$M/C\ PROD = 1.92\ Tons/hr$$

REAM WEIGHT CALCULATION

Let:

Paper Size = 40 cm x 60 cm

GSM = 50

No. of Sheets = 500

$$REAM\ WEIGHT(in\ kilograms)\ =\ \ PAPER\ SIZE\ \times\ GSM\ \times\ NO.OF\ SHEETS$$

$$REAM\ WEIGHT\ =\ \frac{LENGTH\ \times\ WIDTH \times\ GSM \times\ 500}{10^7}$$

$$REAM\ WEIGHT\ =\ \frac{40 \times\ 60\ \times\ 50 \times\ 500}{10^7}$$

$$REAM\ WEIGHT\ =\ 6kg$$

CALCULATION OF GSM

Let

W = weight of paper in grams = 0.72

L = Length of paper in cm = 10

B = width of paper in cm = 15

$$GSM\ =\ \frac{10000\ \times\ W}{l\ \times\ b}$$

$$GSM\ =\ \frac{10000\ \times\ 0.72}{10\ \times\ 15}$$

$$GSM\ =\ 48$$

THEORETICAL HEAD

$$\boldsymbol{THEORETICAL\ HEAD}\ =\ (\frac{\frac{\boldsymbol{V}}{\boldsymbol{100}}}{\boldsymbol{K}})^{\boldsymbol{2}}$$

Where

V= Spouting Velocity (fpm)

K = constant

HEAD	In of H20	Ft of H20	In of hg	Pressure PSIG
K	1.932	23.184	26.345	53.623

HEAD CALCULATION

$$V^2 = 2gh$$

$$h = \frac{V^2}{2g}$$

Let

H = height of stock in headbox

V = Velocity

G = Acceleration Due to Gravity

SHORTCUT

$$h = \frac{SPEEDXSPEED}{706.32}$$

assume 180 m/min

$$h = \frac{180\ x\ 180}{706.32}$$

=45.87cms

JET VELOCITY CALCULATION

Let

H= 89 cm

$$V = \sqrt{2gh}$$

SHORTCUT

$$V = \sqrt{89 \times 706.32}$$

$$V = 250.72\ M/min$$

RETENTION

$$Retension\ \% = \frac{Net\ Consistency}{Head\ Box\ Consistency} \times 100$$

$$Overall\ Retension\ \% = \frac{Filler\ in\ Sheet}{Filler\ Added\ to\ Furnish} \times 100$$

FLUID VELOCITY

$$Velocity\ = \frac{GPM\ \times\ 0.321}{A}$$

ROLL SPEED

$$\boldsymbol{R}PM = \frac{3.82\,V}{Do}$$

Where

RPM = revolutions per minute

V = Speed (fpm)

Do = Roll Outside Diameter in inches

VACUUM PUMP CAPACITY (CFM)

$$\boldsymbol{PV\ =\ NRT}$$

Where:

P = Absolute Pressure

V = Total Gas Volume, ft^3

n = Weight of Gas, lbs.

T = Absolute Temperature, R

R = gas constant

HEAD BOX FLOW RATE

GPM / inch = S.O. x V x 0.052 x C

V = Spouting velocity (fpm)

S.O. = Slice opening (inches)

C = Orifice coefficient (see table for approximate values)

TYPE	C
NOZZLE	0.95
A	0.75
B	0.70
C	0.60

CALCULATION OF WIRE LENGTH

$$\boldsymbol{Ls = 2I + \frac{\pi}{2}(D1 + D2) + K}$$

Ls = Length of wire (in mm)

l = Distance between center of breast roll to couch roll (in mm)

D1 = Diameter of breast roll (in mm)

D2 = Diameter of couch roll (in mm)

K = A constant, 130 mm for fourdrinier wire part.

L = 12,500 mm

D1 = 450 mm

D2 = 800 mm

$$\boldsymbol{Ls = 2(12,500) + \frac{\pi}{2}(450 + 800) + 130}$$

= 25000 + 1.571 (1250) + 130

= 25000 + 1963.75 + 130

= 27093.75 mm

= 27.093 meter

SIZE PRESS ROLL REVOLUTION PER MINUTE

$$\text{RPM} = \frac{\text{web speed (fpm)}}{\text{3.142 x size press roll diameter}}$$

Maximum 250 rpm

HEAD BOX APPROACH SYSTEM STOCK VELOCITIES

$$\mathrm{V(fps)} = \frac{\text{stock flow (gpm)}}{\text{Pipe radius } (ft^2)} = 0.0007092$$

$$\text{acceptable range} = 7 - 14 \text{ fps}$$

WATER EVAPORATED AT THE DRYER

$$\text{water evaporated at the dryer} = \left(\frac{\text{final dryness}}{initial\ dryness} - 1\right) production$$

Let Final dryness = 95 %

Paper web entering the dryer Section of dryness = 38 %

Production = 1200 kg/hr

$$= \left(\frac{95}{38} - 1\right) 1200$$

= 1.5 x 1200 = 1800 kg

1800 kg water evaporated at dryer Section for produce 1200 kg paper in 1 hour.

b. PIPING AND ACCESSORIES

Pulp Type (5)	Pipe Material	K	Σ
Unbeaten aspen sulfite never dried (2)	Stainless Steel	0.85	1.6
Long fibered kraft never dried CSF = 725 (6) (2)	PVC	0.98	1.85
Long fibered kraft never dried CSF = 725 (6) (2)	Stainless Steel	0.89	1.5
Long fibered kraft never dried CSF = 650 (6) (2)	PVC	0.85	1.9
Long fibered kraft never dried CSF = 550 (6) (2)	PVC	0.75	1.65
Long fibered kraft never dried CSF = 260 (6) (2)	PVC	0.75	1.8
Bleached kraft pine dried and reslurried (6) (2)	PVC	0.79	1.5
Bleached kraft pine dried and reslurried (6) (2)	Stainless Steel	0.59	1.45
Long fibered kraft dried and reslurried (6) (2)	PVC	0.49	1.8
Kraft birch dried and reslurried (6) (2)	PVC	0.69	1.3
Stone groundwood CSF = 114 (2)	PVC	4.0	1.40
Refiner groundwood CSF = 150 (2)	PVC	4.0	1.40
Newsprint broke CSF = 75 (2)	PVC	4.0	1.40
Refiner groundwood (hardboard) (2)	PVC	4.0	1.40
Refiner groundwood (insulating board) (2)	PVC	4.0	1.40
Hardwood NSSC CSF = 620 (2)	PVC	0.59	1.8
Unbleached sulfite (1)	Copper	0.98	1.2
Bleached sulfite (1)	Copper	0.98	1.2
Kraft (1)	Copper	0.98	1.2
Bleached straw (1)	Copper	0.98	1.2
Unbleached straw (1)	Copper	0.98	1.2
Cooked groundwood (1)	Copper	0.75	1.8
Soda (1)	Steel	4.0	1.4

c. MATERIAL HANDLING UNITS

BULK SOLIDS

To size conveyors, actual **Wet Weight** of the wood being moved, as moisture adds significant load:

$$\text{total weight} = \frac{\text{bone dry weight}}{1 - \frac{Moisture\ content\ percentage}{100}}$$

VOLUMETRIC CONVEYOR CAPACITY:

When sizing belt conveyors for wood chips, bulk density is highly variable. Use the lowest density to prevent volumetric choking, and the highest density to check structural motor loads:

$$capacity\left(\frac{tons}{hour}\right) = CROSS\ SECTIONAL\ AREA\ X\ BELT\ SPEED\ X\ BULK$$

APPENDIX:

PROPOSED COMPANY PROFILE

Ream-R-Kable Fiber is a forward-thinking, paper and pulp manufacturing corporation strategically situated across a sprawling industrial footprint in **Barangay Ciudad Real, San Jose del Monte, Bulacan**.

Our brand identity stems from a clever play on the English word **"remarkable."** By blending a *"ream"* (the universal unit of paper measurement) with *"fiber"* (the foundational biological element of our product), our name reflects our core mission: to transform raw, sustainably sourced fiber into remarkable, world-class paper products. Nestled in the booming economic hub of San Jose del Monte, our facility bridges the gap between rural raw material supply chains and fast-paced urban markets.

Our Vision

To be the premier, most technologically advanced, and environmentally 'remarkable' pulp and paper mill in Southeast Asia

Our Mission

- **To Deliver Excellence:** To engineer and manufacture exceptionally high-grade paper and pulp products that consistently exceed the technical and qualitative demands of global printing, packaging, and industrial markets.
- **To Champion Sustainability:** To aggressively minimize our environmental footprint by pioneering advanced water-recycling systems and utilizing waste paper as raw material.
- **To Empower the Community:** To act as an economic engine for the province of Bulacan, specifically creating high-value technical jobs, fostering community development, and practicing strict corporate responsibility.
- **To Drive Innovation:** To perpetually redefine pulping and refining efficiency by integrating modern engineering standards

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